

Sarus Firmware Plan

How Sarus builds its own ArduPilot firmware for Plane (QuadPlane included), Copter and Rover, how a release reaches the Install Firmware page, and what has been done so far.

01 Where it stands

The fork exists. It lives at github.com/LikhitS/ardupilot on the branch `sarus-4.7.1`, which starts from ArduPilot's stable 4.7.1 release. Plane, Copter and Rover 4.7.1 are one and the same commit, so a single branch covers all three.

So far the only change to the firmware itself is its name. It now reports ArduPlane V4.7.1 Sarus-1, and likewise for Copter and Rover. Flight behaviour is identical to ArduPilot 4.7.1. The ArduPlane V prefix stays deliberately: ground stations, Sarus Operation Planner included, use it to recognise the vehicle type, and if it were missing the program could treat a QuadPlane as a Copter and offer the wrong flight modes.

GitHub's cloud machines build the firmware for seven board families: CubeOrange, CubeOrangePlus, Pixhawk6X, Pixhawk6C, Pixhawk1, fmuv3 and MatekH743. Every build is checked for the Sarus name. The same run produces the Windows simulator, so the Sarus firmware can be flown in the simulator by the ground-station test harness. It has been: QuadPlane, Copter and Rover flights, the parameter editor checks, shortcuts under load and a sweep of every QuadPlane parameter all passed on the Sarus build.

02 From a release to the Install Firmware page

Tag sarus-v4.7.1-S1 pushed to the fork	Cloud build all boards plus the Windows simulator	Release files named Sarus- Plane-4.7.1-S1- CubeOrange.apj and so on	Firmware list manifest.json on the sarus-manifest branch
Sarus Planner			

reads it at start-up next
to ArduPilot's list

The firmware list is made by ArduPilot's own manifest script, unchanged, so board IDs, USB IDs and bootloader names are exactly the ones ArduPilot publishes. We compared six entries field by field with ArduPilot's official list and found no difference except the download address. When a board matches both an ArduPilot and a Sarus build, Install Firmware opens the selection window Mission Planner already has, and the Sarus files are easy to spot by name. No new buttons were needed.

The program loads the list when it starts. A new Sarus release therefore appears the next time Sarus Operation Planner is opened.

03 Next steps

With the simulator tests passed, the first release, `sarus-v4.7.1-S1`, is ready to be tagged. That needs your approval, because it publishes files. After it, your team flashes the Sarus build on real boards and runs the Field Test Checklist, starting with this identity-only version so that any problem cannot come from a behaviour change.

Behaviour changes come only after that, and one per release. The parameter sweeps have already suggested where to start. `EK3_BCN_DELAY`, `EK3_FLOW_DELAY` and `LAND_THR_SLEW` are stored in 8 bits and stop at 127 although their documentation allows more. `Q_LOIT_ACC_MAX_M` and `LOIT_ACC_MAX_M` are accepted and then limited inside the firmware. Your flight team will find others. Each change gets its own commit, simulator results from before and after, an entry in `SARUS.md` and your sign-off.

SAFETY

Many of ArduPilot's limits protect the aircraft: angle, acceleration and throttle slew limits among them. Removing one moves that protection to the pilot and the tuning. That is the reason for going one limit at a time, with evidence, and never in the first release.

Later, the build can grow from seven board families to everything ArduPilot supports, about 1,100 firmware files per release. That is more than one GitHub release can hold, so it would become one release per vehicle type.

04 Decisions taken

Boards	The seven families above first, then every board Mission Planner offers
Install Firmware	Sarus builds listed together with ArduPilot's, with no new buttons
Repository	Public fork under LikhitS for now, to move to a company account later
Version names	ArduPilot version plus a Sarus build number, e.g. 4.7.1-S1, so the base is always visible
Release types	Stable (sarus-v4.7.1-S1) and beta (sarus-v4.7.1-S1-beta)

05 Licensing

ArduPilot is licensed under the GPL version 3. Sarus firmware may be sold or supplied with aircraft, provided that anyone who receives it can get the complete source of the build they received. The public fork does that. Saying the firmware is based on ArduPilot is fine; suggesting ArduPilot endorses it is not.